

Towards Automated Generation of Benchmark Instances with Diverse Solver Performance

Tianchen Wu, Ian Miguel, Nguyen Dang



University of
St Andrews

What is our aim?

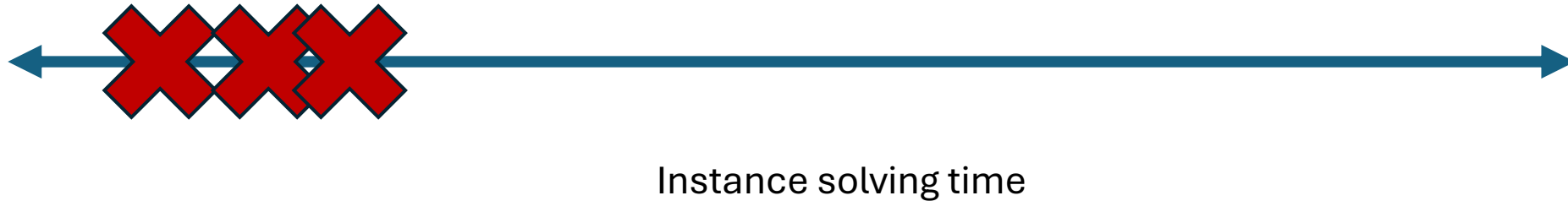
Create benchmarks of combinatorial optimisation problems that have
a large *quantity of diverse* instances *

* Automatically because hand crafting these is hard and arduous

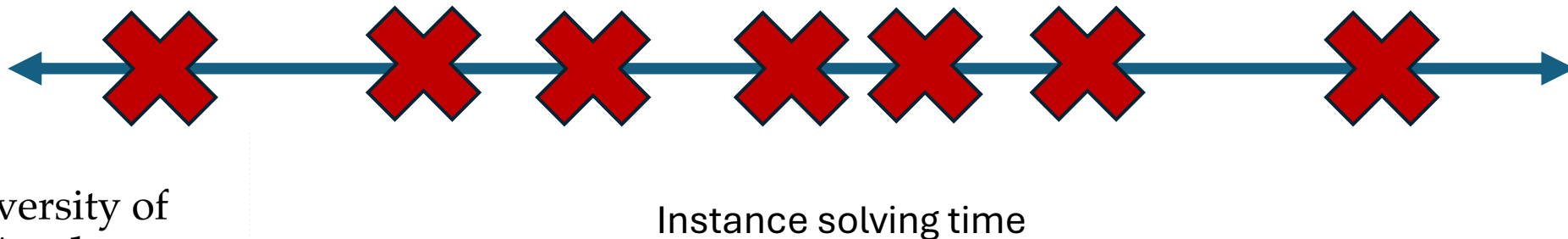


What are Diverse Instances?

- Looking at the time taken for some solver to solve the problem instance, i.e. *performance* space



But also:



How do we define “diversity”?

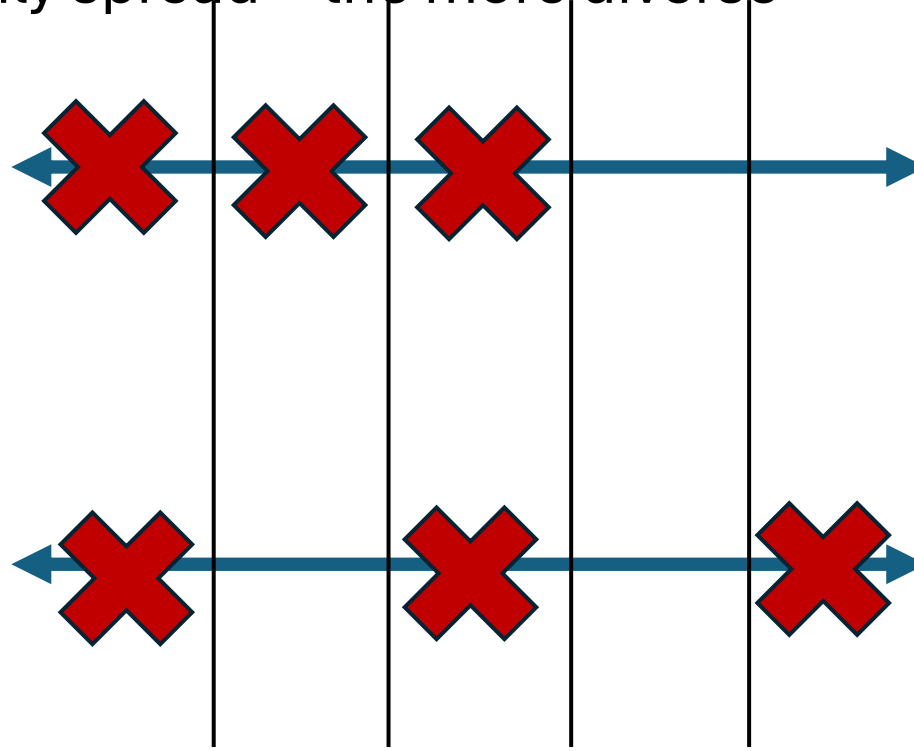
- We want good *coverage* and *spread*
- We also want *yield*
 - a signal shouldn't win because it generates one in each bin and stops
- Measure will need to balance coverage and yield



Defining the measures

- **Shannon entropy**

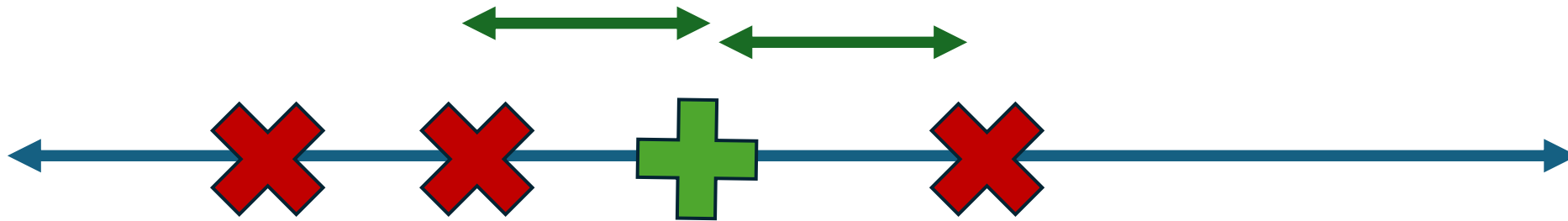
- Shannon Entropy measures how evenly spread a dataset is in discrete terms
- The more evenly spread = the more diverse



Defining the measures

- **Maximising Distance to Closest Neighbour**

- Measure the distance of the new instance to its closest neighbours, and maximise this distance

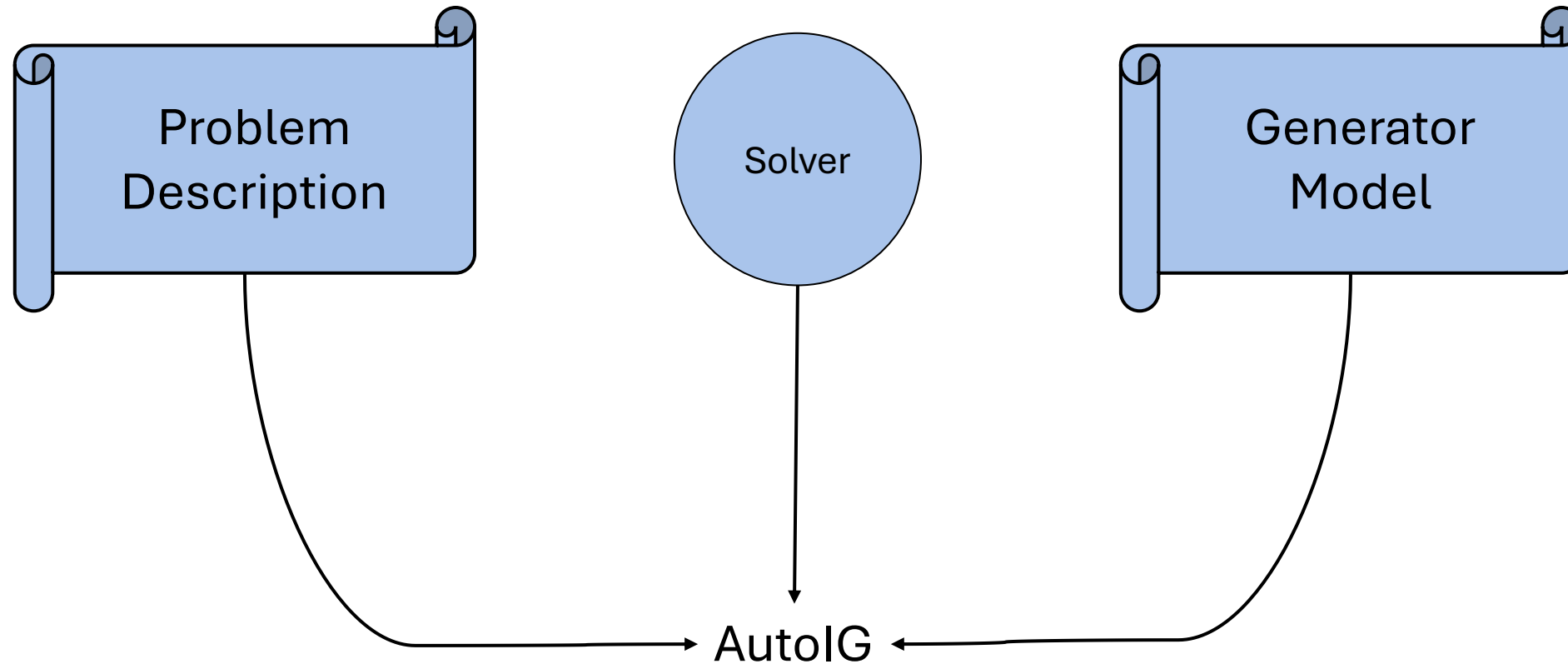


AutoIG

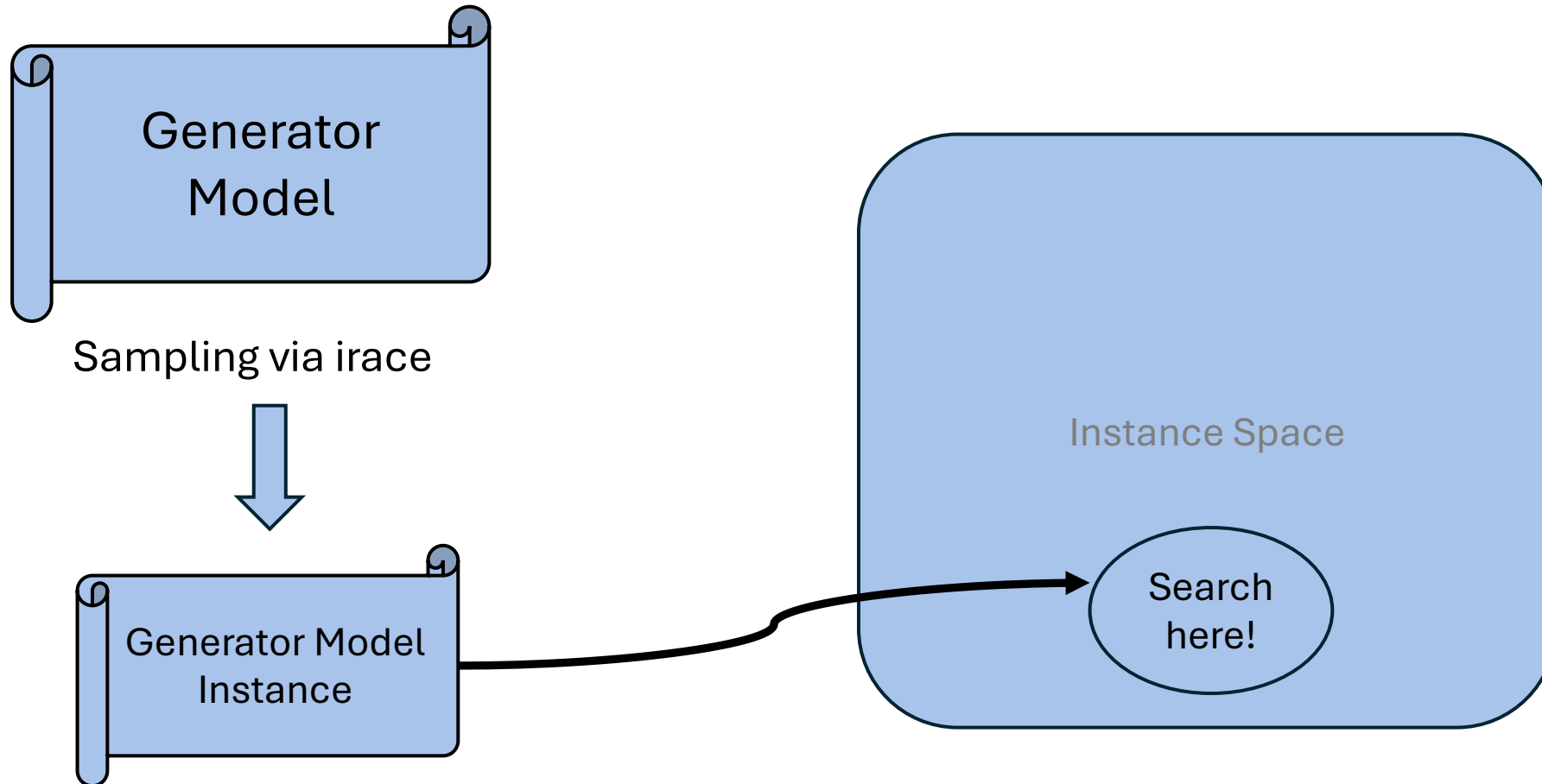
- Constraints programming based instance generator



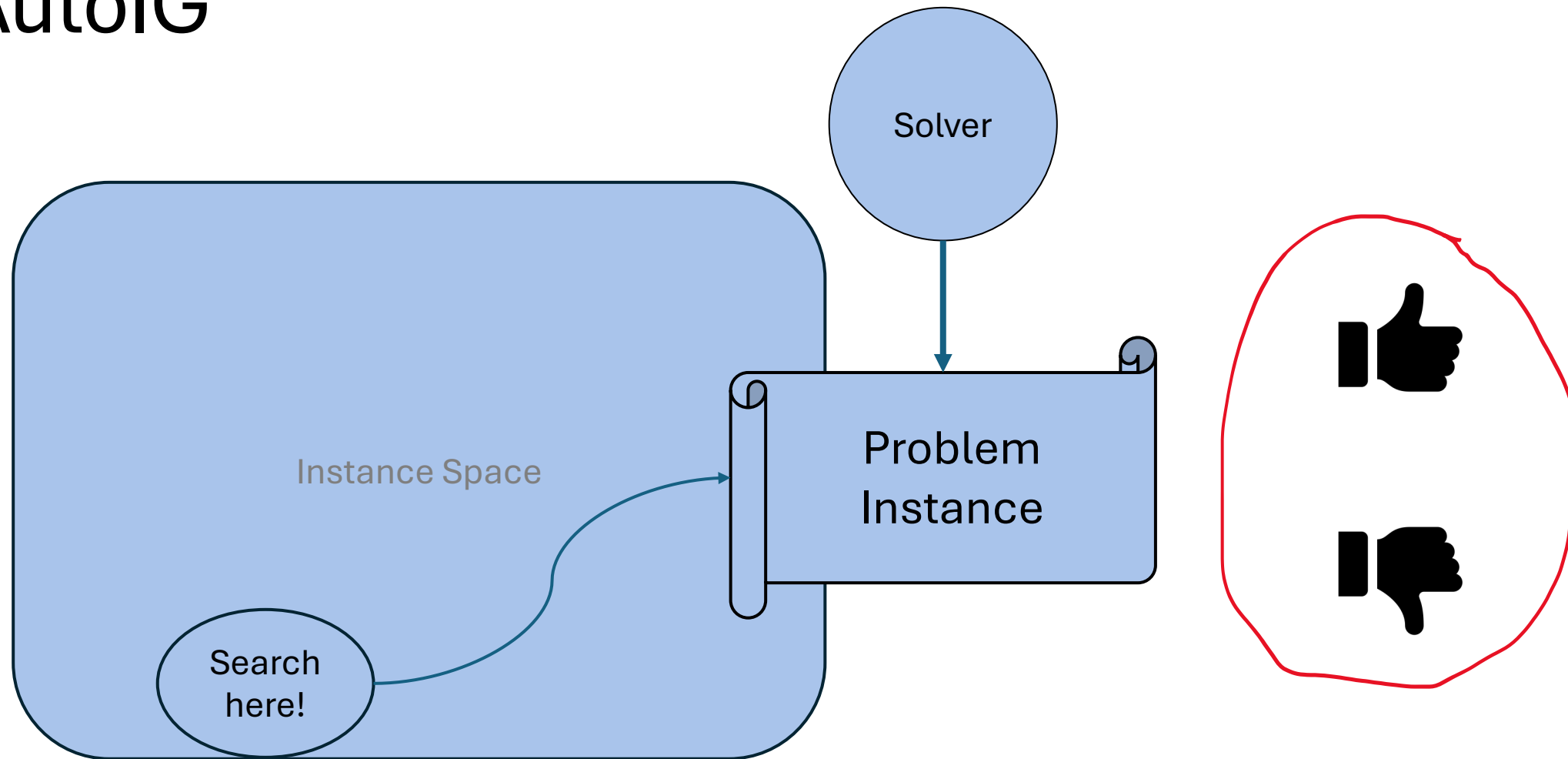
AutoIG



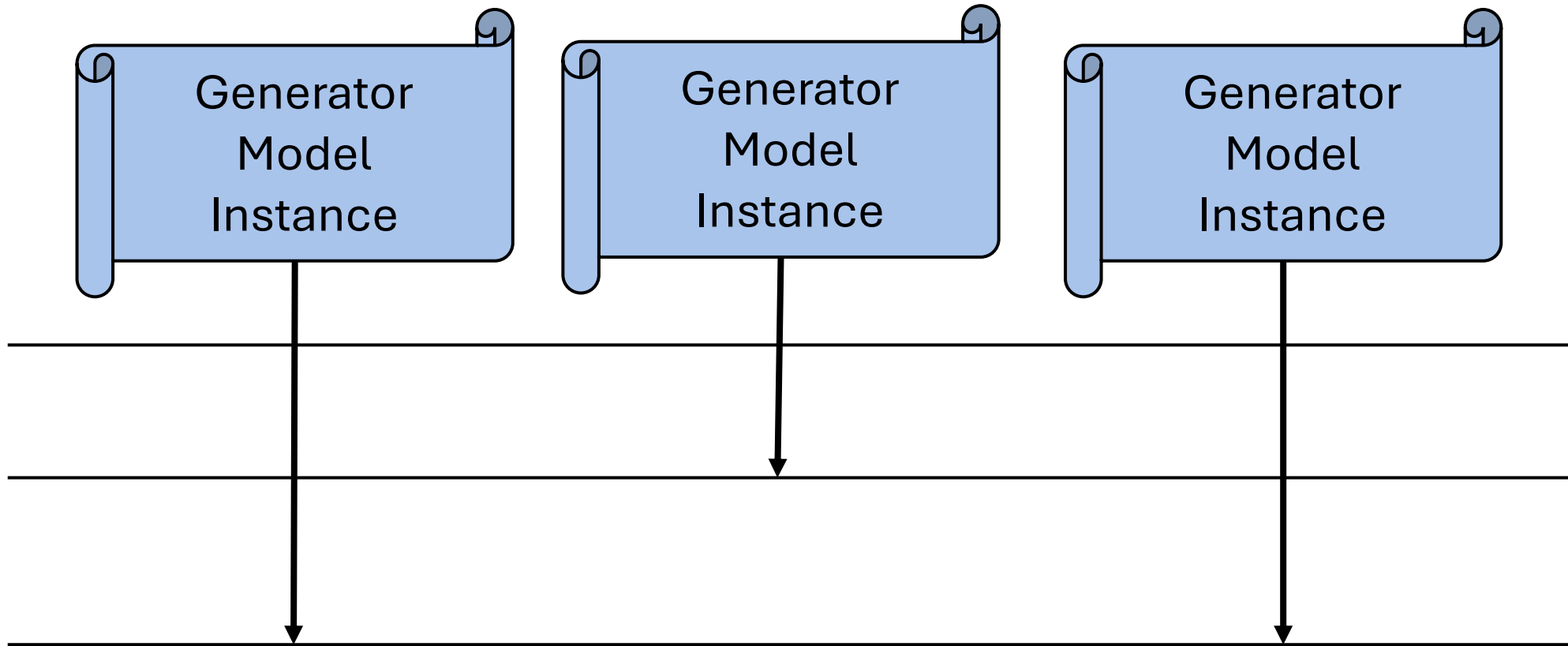
AutoIG



AutoIG

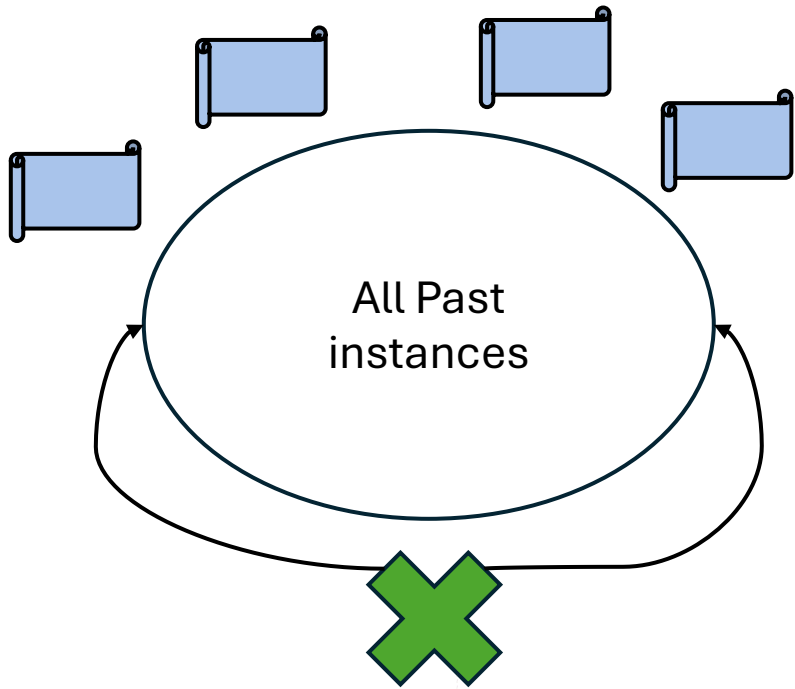


What does this signal do?



Scoring generated instances

Global Method

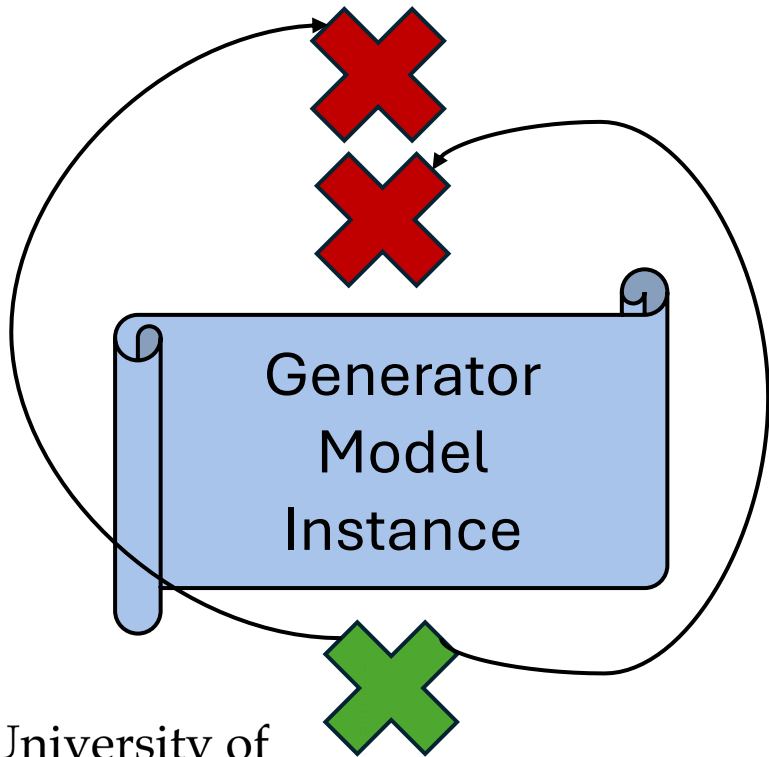


- Shannon Entropy
- Maximising Distance to Closest Neighbours



Scoring generated instances

Local Method

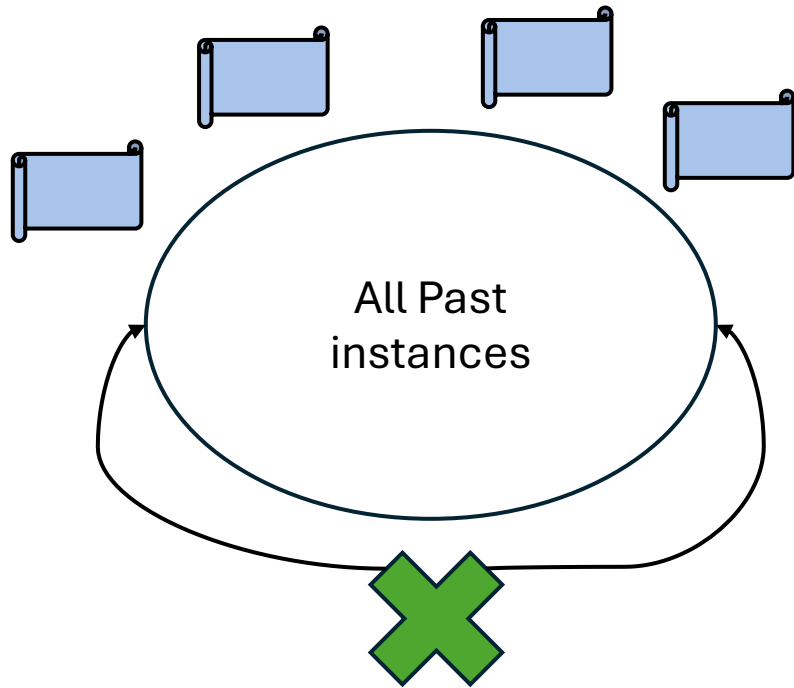


- Shannon Entropy

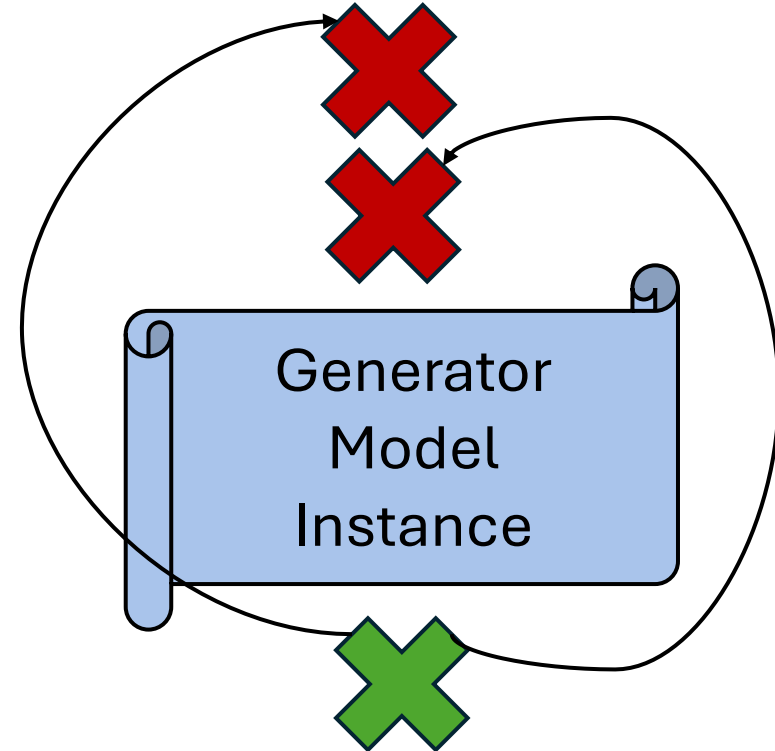


Scoring generated instances

Global Method



Local Method



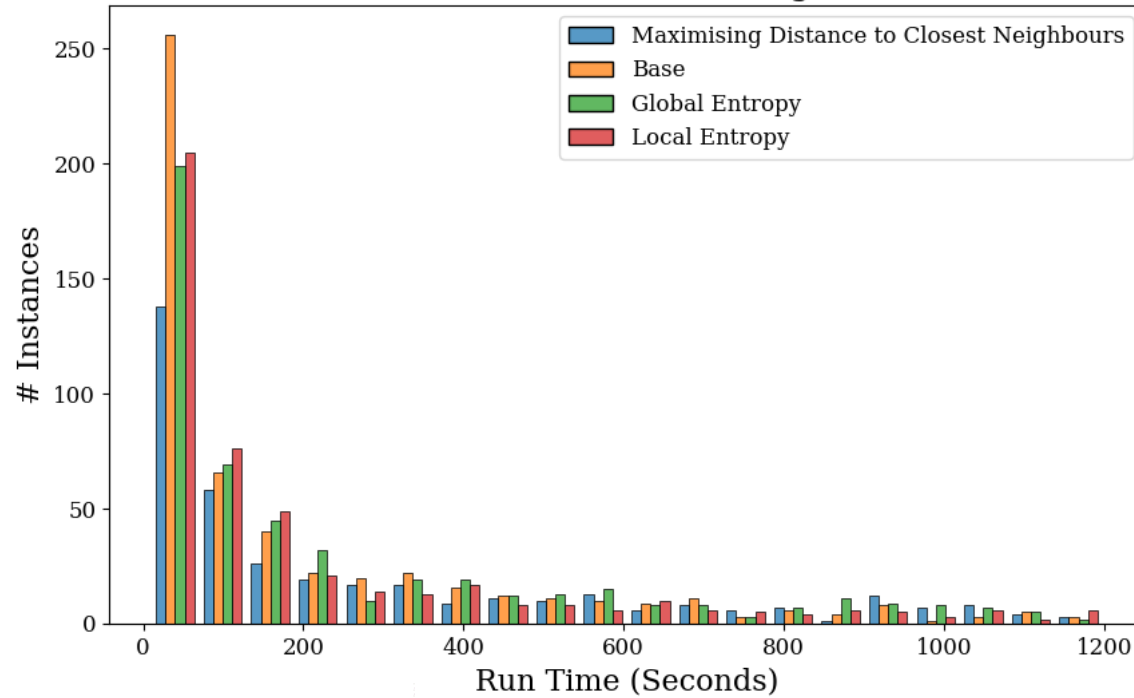
Evaluation and Experimental Setup

- We use two solvers chuffed and OR-Tools CP-SAT (single core)
- Running on two problem classes
 - Multi-agent Collaborative Construction problem (MACC) and Lot-Sizing
- Give each experiment an evaluation budget of 2000
 - 2000 evaluations of generator model instances
- Want instances that take 10 seconds to 1200 seconds to solve

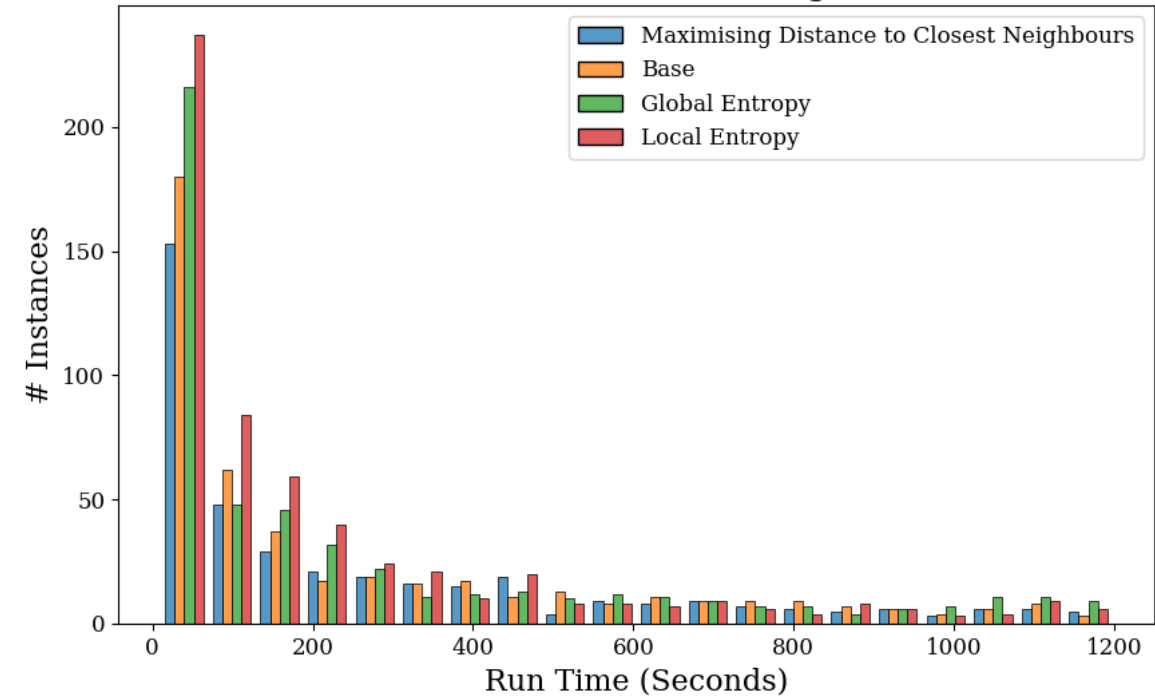


Results – Lot-Sizing

Chuffed on Lot-Sizing

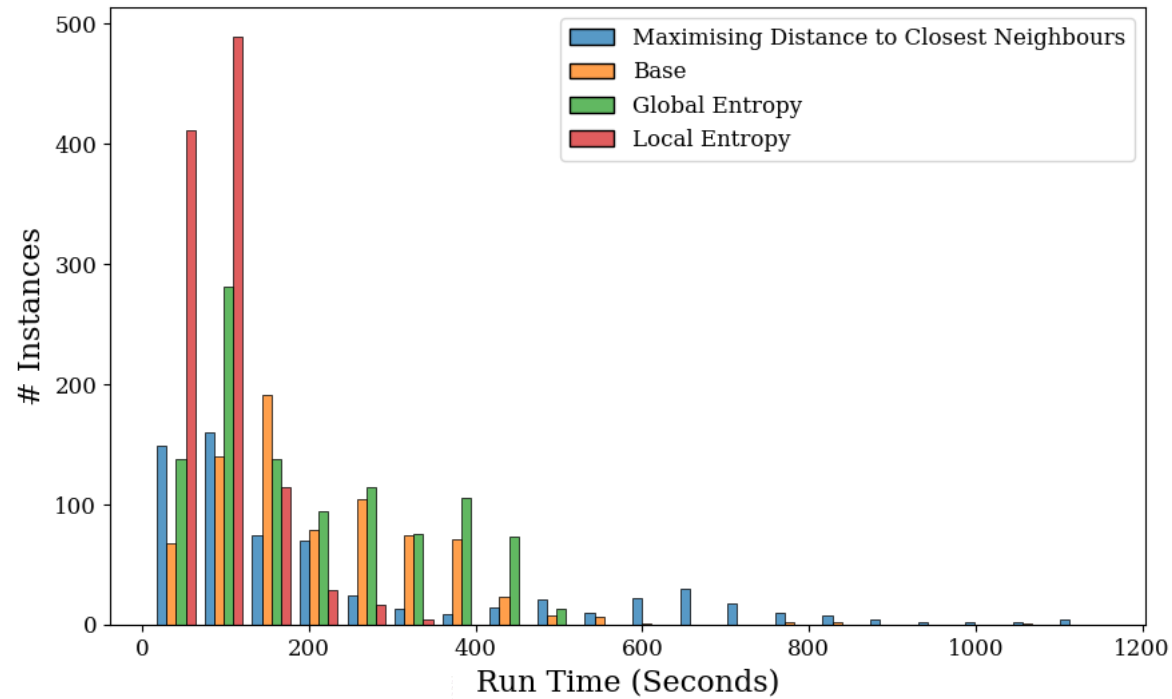


OR-Tools on Lot-Sizing

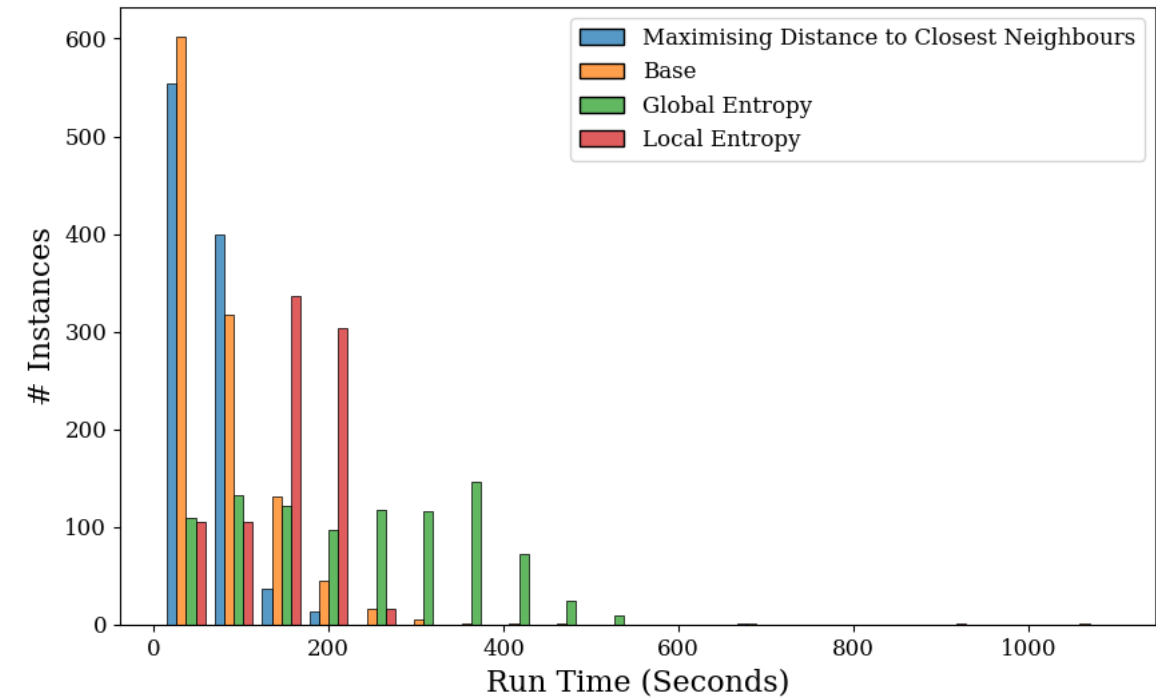


Results – MACC

Chuffed on MACC



OR-Tools on MACC



Post-hoc measures of diversity

Scaled Normalised Shannon Entropy

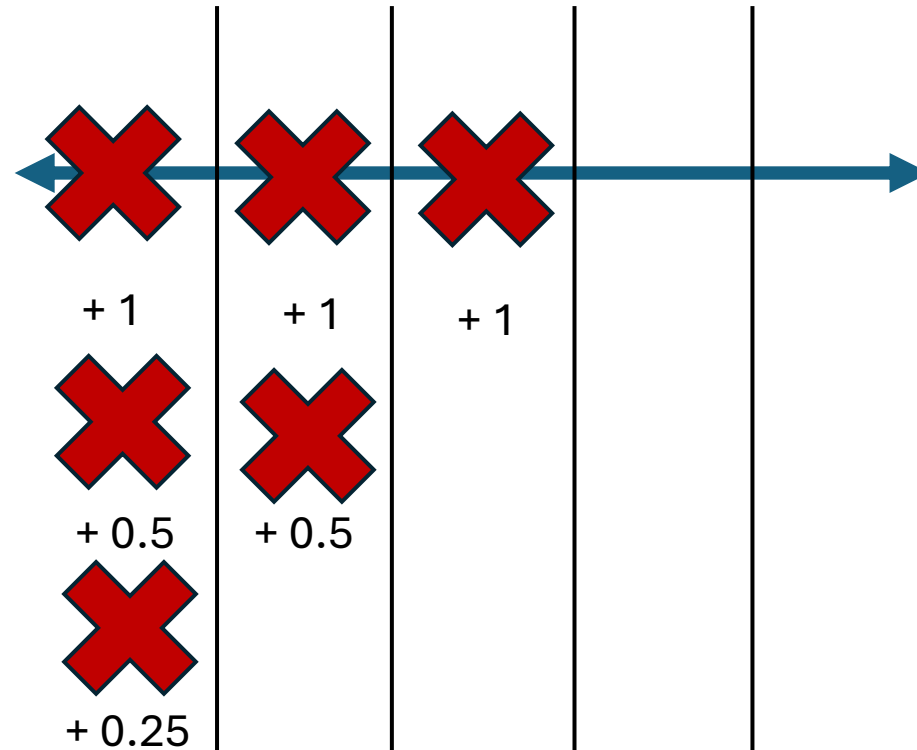
- Using same concept of even distribution
- Scale using a logarithmic multiplier on the number of accepted instances generated
- Does not directly enforce coverage



Post-hoc measures of diversity

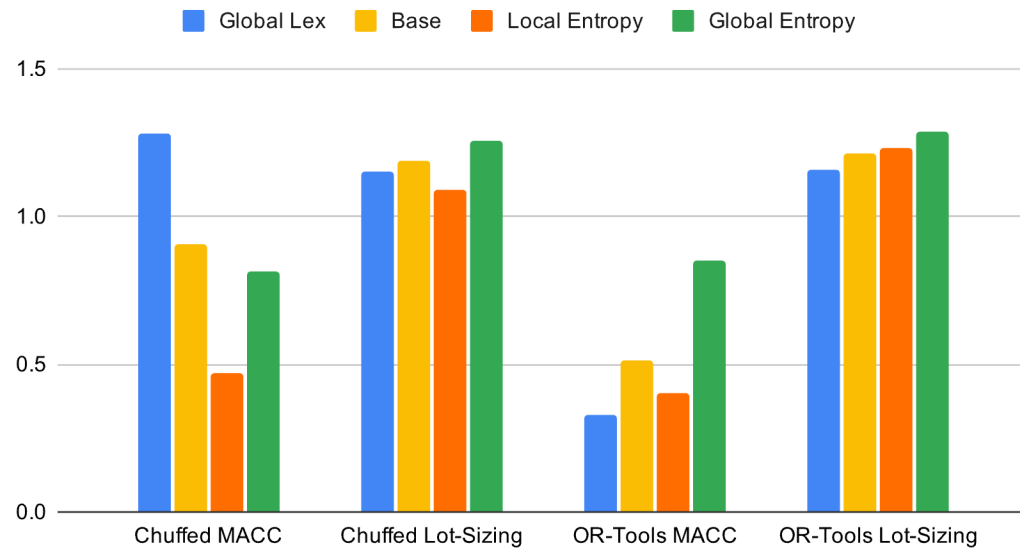
Softened Lexicographic Coverage Score

- Try to prioritise coverage more in the score

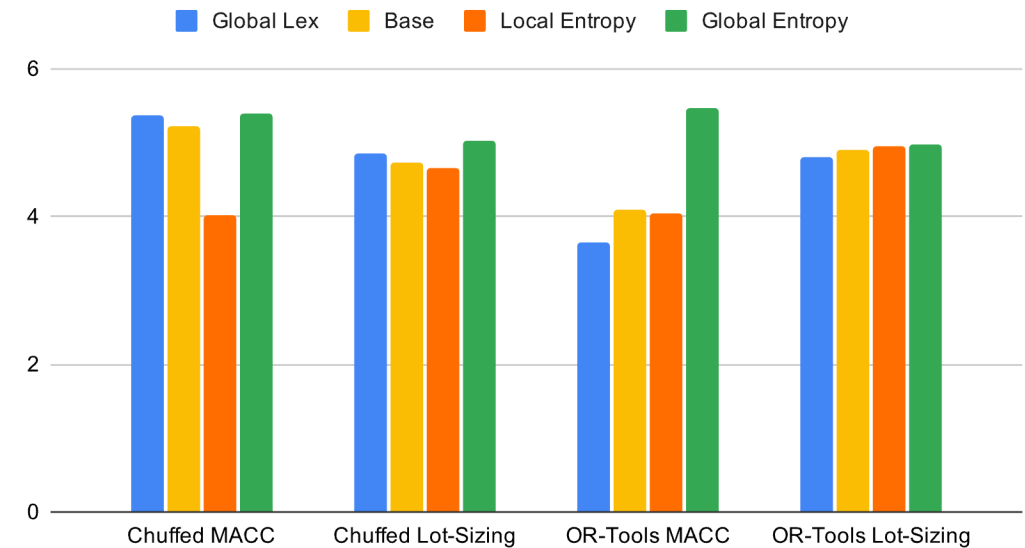


Results of post-hoc analysis

Softened Lexicographic Ordering



Scaled Normalised Shannon Entropy



What's next?

- Running more experiments
- Define diversity metrics that combine coverage and yield more clearly, but also *distance*
- Long term: combine this with feature-based diversity

Thank you



University of
St Andrews